

HeatGeo: Corpus-Level Heat-Kernel Geometry Distillation for Compact Text Embedding Models

Abstract

Large text embedding models provide strong semantic representations but are expensive to deploy. Knowledge distillation has become a practical way to compress these models into compact encoders. Existing embedding distillation methods mostly transfer teacher knowledge through pointwise embedding alignment, pairwise mini-batch geometry, token-level representation matching, or multi-faceted loss combinations. However, these objectives treat the teacher embedding space as a collection of local vectors or mini-batch relations, while real embedding models are deployed for corpus-level retrieval, clustering, and nearest-neighbor search. This creates a gap: current methods do not explicitly distill the global semantic manifold induced by the teacher over the training corpus.

We propose **HeatGeo**, a corpus-level geometry distillation framework that transfers the teacher’s semantic diffusion process rather than only its individual embeddings. HeatGeo first constructs a teacher-induced mutual k -nearest-neighbor graph over the training corpus. It then derives multi-scale heat-kernel diffusion distributions from random walks on this graph. During training, the student is optimized to match the teacher’s diffusion distribution over sampled candidate neighborhoods, enabling it to preserve local neighbors, semantic communities, and multi-hop manifold structure. To further anchor global topology, HeatGeo introduces a spectral manifold objective based on graph Laplacian eigenvectors. The resulting framework directly optimizes the retrieval geometry that embedding models are used for, without requiring teacher hidden states or modifying the student architecture. We hypothesize that corpus-level heat diffusion provides a stronger and more task-aligned supervision signal than static mini-batch relational objectives, enabling compact students to outperform strong cross-tokenizer embedding distillation baselines.

1 Motivation

Text embedding models are central to semantic retrieval, sentence similarity, clustering, reranking, and retrieval-augmented generation. Recent large embedding models such as LLM-based encoders achieve strong performance, but their parameter size and inference cost limit deployment in resource-constrained settings. A common solution is knowledge distillation, where a compact student encoder learns from a stronger teacher embedding model.

Most existing embedding distillation methods fall into one of four categories. First, pointwise methods align each student embedding with its corresponding teacher embedding. Second, relational methods preserve pairwise similarity matrices within a mini-batch. Third, cross-tokenizer methods align token-level hidden states, attention maps, or output distributions between teacher and student. Fourth, multi-faceted methods combine several losses, such as pointwise, contrastive, and pairwise objectives.

Despite their effectiveness, these methods share a hidden limitation: they mostly distill *static local geometry*. A pointwise objective only teaches the student where each sample should lie. A mini-batch pairwise objective only teaches the student the similarity structure among samples present in the same batch. Token-level objectives transfer fine-grained structural information, but they do not directly optimize the

corpus-level nearest-neighbor structure that determines retrieval performance. In practice, embedding models are evaluated by how well they organize an entire corpus into meaningful neighborhoods and semantic communities.

This motivates the central question of this proposal:

Can we distill a large embedding teacher by matching its corpus-level semantic diffusion process, rather than only matching individual vectors or mini-batch relations?

We answer this question with **HeatGeo**, a method that views the teacher embedding space as a semantic graph and distills the heat diffusion dynamics on that graph into the student.

2 Research Gap

2.1 Limitations of Pointwise and Pairwise Distillation

Pointwise embedding distillation minimizes a distance such as cosine distance or mean squared error between student and teacher embeddings:

$$\mathcal{L}_{\text{point}} = \frac{1}{N} \sum_{i=1}^N (1 - \cos(W s_i, t_i)), \quad (1)$$

where s_i and t_i denote student and teacher embeddings for input x_i , and W is an optional projection matrix.

Although simple and efficient, pointwise alignment can over-constrain a compact student. The student may fail to reproduce the teacher’s absolute embedding coordinates due to capacity mismatch, dimensional mismatch, or architectural differences. More importantly, retrieval does not require exact coordinate matching; it requires preserving neighborhoods and ranking structure.

Pairwise relational distillation improves upon pointwise alignment by preserving mini-batch similarity matrices:

$$\mathcal{L}_{\text{pair}} = \|G_B^S - G_B^T\|_F^2, \quad (2)$$

where G_B^S and G_B^T are student and teacher cosine similarity matrices within a mini-batch B . However, mini-batch geometry is only a local and noisy approximation of the full corpus geometry. Two semantically connected samples may rarely appear in the same batch, and many multi-hop relations are invisible to batch-level objectives.

2.2 Limitations of Token-Level Cross-Tokenizer Distillation

Recent cross-tokenizer distillation methods address the mismatch between teacher and student tokenizers through token mapping, optimal transport, hidden-state alignment, or attention-pattern alignment. These methods are useful because tokenizer mismatch can prevent direct token-wise supervision. However, token-level alignment optimizes intermediate representations, not necessarily the final corpus-level retrieval geometry.

Moreover, token-level methods may introduce additional cost because they often require teacher hidden states, attention maps, or parallel teacher-student forward passes. For large LLM-based embedding teachers, storing or computing such dense internal signals can be expensive.

2.3 Limitations of Multi-Faceted Distillation

Multi-faceted distillation methods recognize that a teacher embedding space contains several types of knowledge, including local identity, global separability, and pairwise relational geometry. However, they still operate primarily at the sequence or mini-batch level. Even when multiple geometric facets are combined, the supervision signal remains a static snapshot of local relations rather than a global semantic process over the training corpus.

2.4 Core Gap

The core gap is therefore:

Existing embedding distillation methods preserve pointwise, token-level, or mini-batch relational knowledge, but they do not explicitly preserve the teacher-induced corpus-level semantic manifold.

HeatGeo addresses this gap by distilling heat-kernel diffusion distributions over a teacher semantic graph.

3 Proposed Method

3.1 Problem Setup

Let

$$\mathcal{D} = x_i * i = 1^N \quad (3)$$

be the training corpus. A large teacher encoder T maps each input x_i to a teacher embedding:

$$t_i = T(x_i) \in \mathbb{R}^{d_T}. \quad (4)$$

A compact student encoder $S * \theta$ maps the same input to a student embedding:

$$s_i = S_\theta(x_i) \in \mathbb{R}^{d_S}. \quad (5)$$

The goal is to train S_θ such that its embedding space preserves the semantic organization induced by the teacher, especially nearest-neighbor structure, semantic communities, and multi-hop manifold relationships.

3.2 Teacher Semantic Graph Construction

We first encode the entire training corpus using the teacher model and build a teacher semantic graph. For each sample x_i , we retrieve its top- k nearest neighbors under teacher cosine similarity:

$$\mathcal{N}_T^k(i) = \text{TopK}_{j \neq i} \cos(t_i, t_j). \quad (6)$$

To reduce noisy one-directional edges and hubness, we use a mutual k NN graph:

$$j \in \mathcal{M}_T^k(i) \iff j \in \mathcal{N}_T^k(i) \text{ and } i \in \mathcal{N}_T^k(j). \quad (7)$$

The weighted adjacency matrix $W^T \in \mathbb{R}^{N \times N}$ is defined as:

$$W_{ij}^T = \begin{cases} \exp(\cos(t_i, t_j)/\tau_g), & j \in \mathcal{M}_T^k(i), \\ 0, & \text{otherwise,} \end{cases} \quad (8)$$

where τ_g is a graph temperature controlling edge sharpness.

Let D^T be the diagonal degree matrix:

$$D_{ii}^T = \sum_{j=1}^N W_{ij}^T. \quad (9)$$

We then define the teacher transition matrix:

$$P^T = (D^T)^{-1} W^T. \quad (10)$$

Each row P_i^T defines a one-step semantic random-walk distribution from sample x_i to its teacher-induced neighbors.

3.3 Multi-Scale Heat Diffusion

A one-step random walk captures local nearest-neighbor relations, but embedding spaces also contain higher-order semantic structure. For example, two sentences may not be direct nearest neighbors, yet they may belong to the same semantic region through intermediate samples. We model such structure using multi-step diffusion:

$$P_r^T = (P^T)^r, \quad (11)$$

where $r \in \mathcal{R}$ is a diffusion scale, such as:

$$\mathcal{R} = 1, 2, 4. \quad (12)$$

Each scale captures a different level of semantic structure:

- $r = 1$: local nearest-neighbor structure;
- $r = 2$: local semantic region;
- $r = 4$: higher-order community structure.

This multi-scale diffusion is analogous to heat propagation over a graph, where semantic mass gradually spreads from an anchor sample to its neighboring semantic region.

3.4 Candidate Set Sampling

Computing a full distribution over all N samples is expensive. Instead, for each anchor x_i , we construct a candidate set:

$$C_i = C_i^+ \cup C_i^h \cup C_i^r, \quad (13)$$

where:

- C_i^+ contains top diffusion neighbors under P_r^T ;
- C_i^h contains teacher hard negatives with moderate teacher similarity but low diffusion probability;
- C_i^r contains random negatives.

This design makes the learning problem non-trivial: the student must assign high probability to diffusion-consistent semantic neighbors while separating hard negatives.

3.5 Teacher Diffusion Target

For each scale r , we restrict the teacher diffusion distribution to the candidate set C_i :

$$p_{i,r}^T(j) = \frac{(P_r^T) * ij}{\sum_{j' \in C_i} (P_r^T)_{ij'}}, \quad j \in C_i. \quad (14)$$

This distribution serves as a soft target. Unlike binary positive/negative labels, it encodes graded semantic relatedness derived from the teacher’s corpus-level manifold.

3.6 Student Neighborhood Distribution

Given student embeddings, we compute the student distribution over the same candidate set:

$$p_i^S(j) = \frac{\exp(\cos(s_i, s_j)/\tau_s)}{\sum_{j' \in C_i} \exp(\cos(s_i, s_{j'})/\tau_s)}, \quad j \in C_i, \quad (15)$$

where τ_s is the student softmax temperature.

3.7 Diffusion Distillation Loss

The main HeatGeo objective matches the student neighborhood distribution to the teacher heat diffusion distribution:

$$\mathcal{L}_{\text{diff}} = \frac{1}{B} \sum_{i \in B} \sum_{r \in \mathcal{R}} \omega_r \text{KL}(p_{i,r}^T(\cdot) | p_i^S(\cdot)), \quad (16)$$

where B is a mini-batch of anchors and ω_r is the scale weight.

A typical setting is:

$$\omega_1 > \omega_2 > \omega_4, \quad (17)$$

because local neighborhoods should remain sharper than long-range diffusion.

3.8 Spectral Manifold Anchoring

While diffusion matching preserves local and multi-hop neighborhoods, we further encourage the student to preserve global corpus topology using spectral coordinates from the teacher graph.

We define the normalized teacher graph Laplacian:

$$L^T = I - (D^T)^{-1/2} W^T (D^T)^{-1/2}. \quad (18)$$

Let $U^T \in \mathbb{R}^{N \times m}$ be the matrix of the first m non-trivial eigenvectors of L^T . The row $u_i^T = U^T[i]$ gives a low-dimensional spectral coordinate for x_i , capturing its global position in the teacher manifold.

We project the student embedding into the spectral coordinate space:

$$z_i^S = W_{\text{spec}} s_i, \quad (19)$$

where $W_{\text{spec}} \in \mathbb{R}^{m \times d_S}$ is a learnable projection.

The spectral loss is:

$$\mathcal{L}_{\text{spec}} = \frac{1}{B} \sum_{i \in B} \|z_i^S - u_i^T\|_2^2. \quad (20)$$

This objective prevents the student from only fitting local neighborhoods while ignoring global semantic communities.

3.9 Lightweight Anchor Regularization

To stabilize training, we optionally include a weak pointwise teacher anchor:

$$\mathcal{L}_{\text{anchor}} = \frac{1}{B} \sum_{i \in B} (1 - \cos(W_a s_i, t_i)), \quad (21)$$

where W_a maps the student embedding to the teacher dimension if needed. This term is not the main contribution; it serves as a boundary condition to keep the student space semantically aligned with the teacher.

3.10 Task or Contrastive Loss

When supervised labels are available, we include a task loss:

$$\mathcal{L}_{\text{task}} = \text{CE}(f(s_i), y_i). \quad (22)$$

For unsupervised distillation, we use a SimCSE-style dropout contrastive loss:

$$\mathcal{L}_{\text{con}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\cos(s_i, s_i^+)/\tau_c)}{\sum_{j=1}^B \exp(\cos(s_i, s_j^+)/\tau_c)}. \quad (23)$$

3.11 Final Objective

The final HeatGeo training objective is:

$$\mathcal{L}_{\text{HeatGeo}} = \mathcal{L}_{\text{task/con}} + \lambda_{\text{diff}} \mathcal{L}_{\text{diff}} + \lambda_{\text{spec}} \mathcal{L}_{\text{spec}} + \lambda_{\text{anchor}} \mathcal{L}_{\text{anchor}}. \quad (24)$$

In our main design, the expected relative importance is:

$$\lambda_{\text{diff}} > \lambda_{\text{spec}} > \lambda_{\text{anchor}}. \quad (25)$$

3.12 Curriculum over Diffusion Scales

Directly matching long-range diffusion early in training may over-smooth the student space. Therefore, we use a curriculum:

- Epoch 1: train with $\mathcal{L}_{\text{anchor}}$ and P^1 diffusion;
- Epoch 2: add P^2 diffusion;
- Epoch 3 onward: add P^4 diffusion and spectral anchoring.

This curriculum reflects the natural learning process: the student first learns local neighborhoods, then semantic regions, and finally global manifold structure.

Algorithm 1 HeatGeo Training

Require: Training corpus $\mathcal{D}$, teacher encoder T , student encoder S_θ , neighbor size k , diffusion scales $\mathcal{R}$.

- 1: Encode all samples with teacher: $t_i = T(x_i)$.
 - 2: Build teacher mutual k NN graph W^T .
 - 3: Compute transition matrix $P^T = (D^T)^{-1}W^T$.
 - 4: Precompute diffusion neighbors and scores from $(P^T)^r$ for $r \in \mathcal{R}$.
 - 5: Compute spectral coordinates U^T from graph Laplacian L^T .
 - 6: **for** each training step **do**
 - 7: Sample anchor batch B .
 - 8: **for** each anchor $i \in B$ **do**
 - 9: Construct candidate set $C_i = C_i^+ \cup C_i^h \cup C_i^r$.
 - 10: Form teacher diffusion target $p_{i,r}^T$ over C_i .
 - 11: Encode anchor and candidates with student.
 - 12: Compute student distribution p_i^S over C_i .
 - 13: **end for**
 - 14: Compute $\mathcal{L} * \text{diff}$, $\mathcal{L} * \text{spec}$, $\mathcal{L} * \text{anchor}$, and $\mathcal{L} * \text{task/con}$.
 - 15: Update θ by minimizing $\mathcal{L} * \text{HeatGeo}$.
 - 16: **end for**
 - 17: **return** Distilled student encoder $S * \theta$.
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Method	Type	Distilled Object	Scope
Pointwise KD	Individual embeddings	Instance-level	Pairwise KD
CTKD	Hidden states / attention	Token-level	Multi-faceted KD
HeatGeo	Semantic heat diffusion	Corpus-level	Similarity matrix
			Mini-batch-level
			Token-level
			Multiple local geometry losses
			Mini-batch-level

Table 1: Conceptual comparison between HeatGeo and existing embedding distillation methods.

4 Why HeatGeo Is Different

HeatGeo differs from existing embedding distillation methods in the object being distilled.

Instead of treating teacher knowledge as isolated vectors or local relations, HeatGeo treats the teacher as defining a semantic diffusion process over the corpus. This provides several advantages:

- **Retrieval-aligned supervision:** The loss directly optimizes neighborhood distributions, which are central to embedding-based retrieval.
- **Multi-hop semantic transfer:** Diffusion captures indirect semantic relations that pairwise mini-batch losses cannot observe.
- **Global topology preservation:** Spectral anchoring transfers corpus-level community structure.
- **Cross-tokenizer robustness:** The main method does not require token-level alignment, making it naturally compatible with heterogeneous teacher-student tokenizers.
- **Architecture preservation:** HeatGeo does not require adding MoE modules or changing the student architecture.

5 Theoretical Intuition

5.1 Diffusion as Manifold-Aware Distillation

The teacher graph W^T can be interpreted as a discrete approximation of the semantic manifold induced by the teacher embedding space. The transition matrix P^T defines a Markov diffusion process over this manifold. Matching P_r^T at multiple scales encourages the student to preserve not only direct neighbors but also higher-order semantic connectivity.

If P^S denotes the analogous transition matrix induced by the student embedding space, then an ideal goal is:

$$P_r^S \approx P_r^T, \quad \forall r \in \mathcal{R}. \quad (26)$$

Because computing P^S over the full corpus at every step is expensive, HeatGeo approximates this objective by matching local candidate distributions p_i^S to precomputed teacher diffusion distributions $p_{i,r}^T$.

5.2 Relation to Heat Kernels

The continuous heat kernel on a manifold is:

$$K_t = \exp(-tL), \quad (27)$$

where L is the Laplace-Beltrami operator. On a graph, heat diffusion can be approximated by powers of a transition matrix or by exponentiating the graph Laplacian. HeatGeo uses multi-step transition powers $(P^T)^r$ as an efficient approximation to heat propagation over the teacher semantic graph.

5.3 Why Multi-Scale Matching Matters

A single diffusion scale may be insufficient:

- small r preserves fine-grained nearest-neighbor relations but may miss semantic communities;
- large r captures global structure but may over-smooth local distinctions.

Therefore, HeatGeo uses multiple scales:

$$\mathcal{R} = 1, 2, 4, \quad (28)$$

balancing local precision and global semantic coherence.